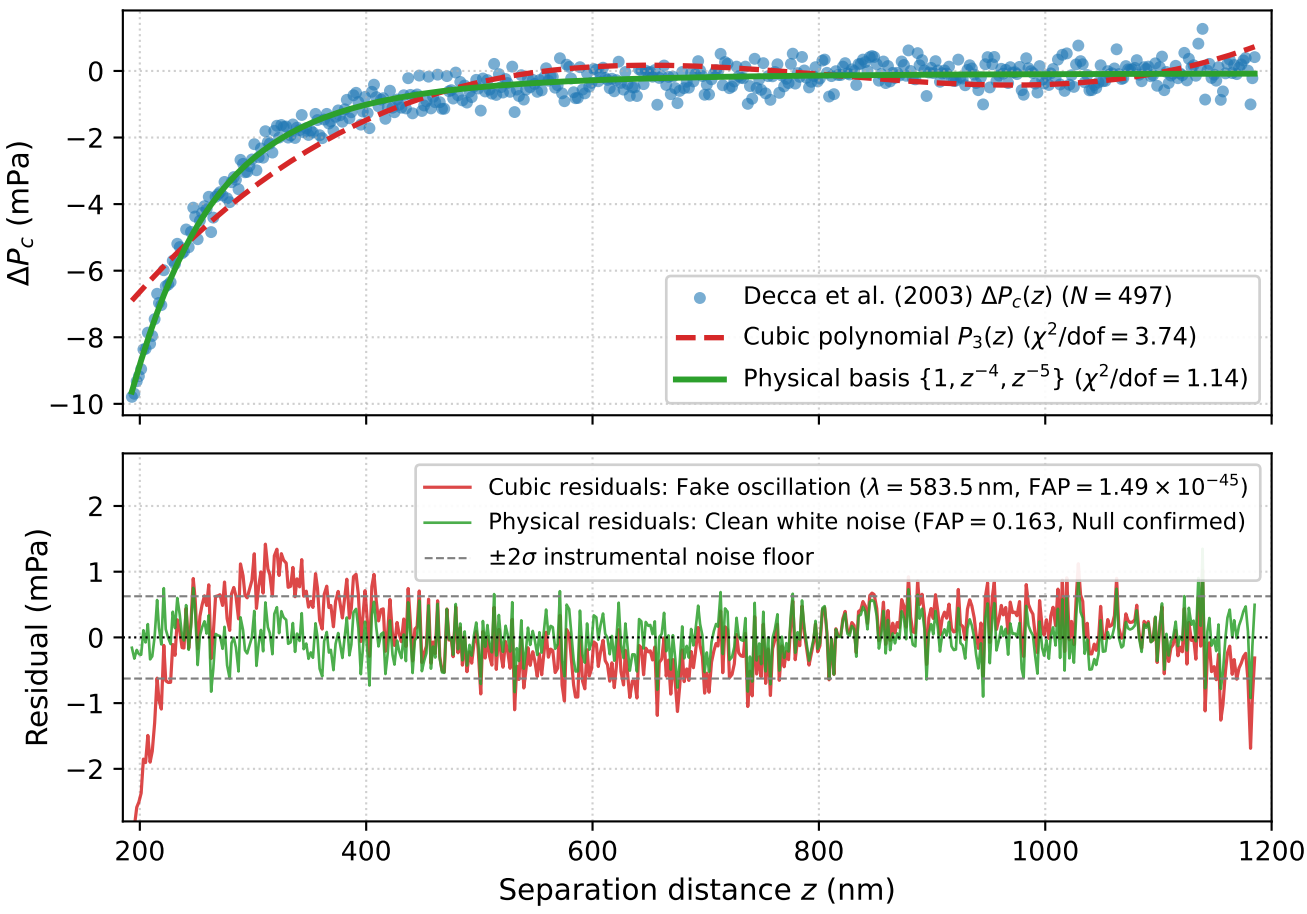
\textbf{Trap 2: Runge Oscillations from Polynomial Detrending on Casimir Data}
ΔP_c (mPa)
Decca et al. (2003) $\Delta P_c(z)$ ($N = 497$)
Cubic polynomial $P_3(z)$ (χ^2/dof = 3.74)
Physical basis $\{1, z^{-4}, z^{-5}\}$ (χ^2/dof = 1.14)
Residual (mPa)
Cubic residuals: Fake oscillation ($\lambda = 583.5$ nm, FAP = 1.49×10^{-45})
Physical residuals: Clean white noise (FAP = 0.163, Null confirmed)
$\pm 2\sigma$ instrumental noise floor
Separation distance z (nm)